Causal inference for marketing

a PyMC-ecosystem cookbook

Contents

This is the second anchor: **geo-lift via synthetic control** — the archetypal "we couldn't randomize" case, and the one MBAs recognise from their own companies. A retailer ran a campaign (a regional TV/out-of-home burst) in **one market (DMA)** and left the rest dark. There is no A/B test; the treated market simply differs from the others in a hundred ways. The questions are the real ones: *did the campaign work, how sure are we, and should we roll it out nationally?* We interrogate it along the same three axes as the uplift anchor. All numbers were executed in PyMC on a simulated 60-week, 30-market panel with a known injected lift (sales in **€000 per week**).

Depth 1 — Estimator and failure modes: why "before vs after" lies

The obstacle is that *everything* moved over the campaign window — an overall trend, seasonality, and a common macro shock all pushed sales up in the post period, in the treated market and everywhere else. Any estimator that attributes the post-launch rise to the campaign is confounded by those common movements. Two tempting naive estimators do exactly that:

- **Naive pre/post** (treated after − treated before): estimates **€24.3k/wk**.
- **Treated − average control**: estimates **€27.1k/wk**.
- **True lift**: **€17.0k/wk.**

Both overstate the effect by roughly **50%**, because they bake in the trend/seasonal rise. The **synthetic control** fixes this by constructing a bespoke counterfactual: a weighted blend of the untreated markets, with weights chosen (on the simplex — non-negative, summing to one) so the blend *tracks the treated market's pre-launch trajectory*. Whatever common forces move the treated market also move its synthetic twin, so subtracting them cancels the confounding. Its estimate is **€15.2k/wk** — close to the truth, and mildly conservative (more on that below).

The pre-launch tracking is the whole game: because the synthetic twin matches the treated market for 40 weeks *before* the campaign — with no lift assumed — its post-launch divergence is credibly the campaign rather than a pre-existing difference. The naive estimators have no such control and simply ride the common trend upward.

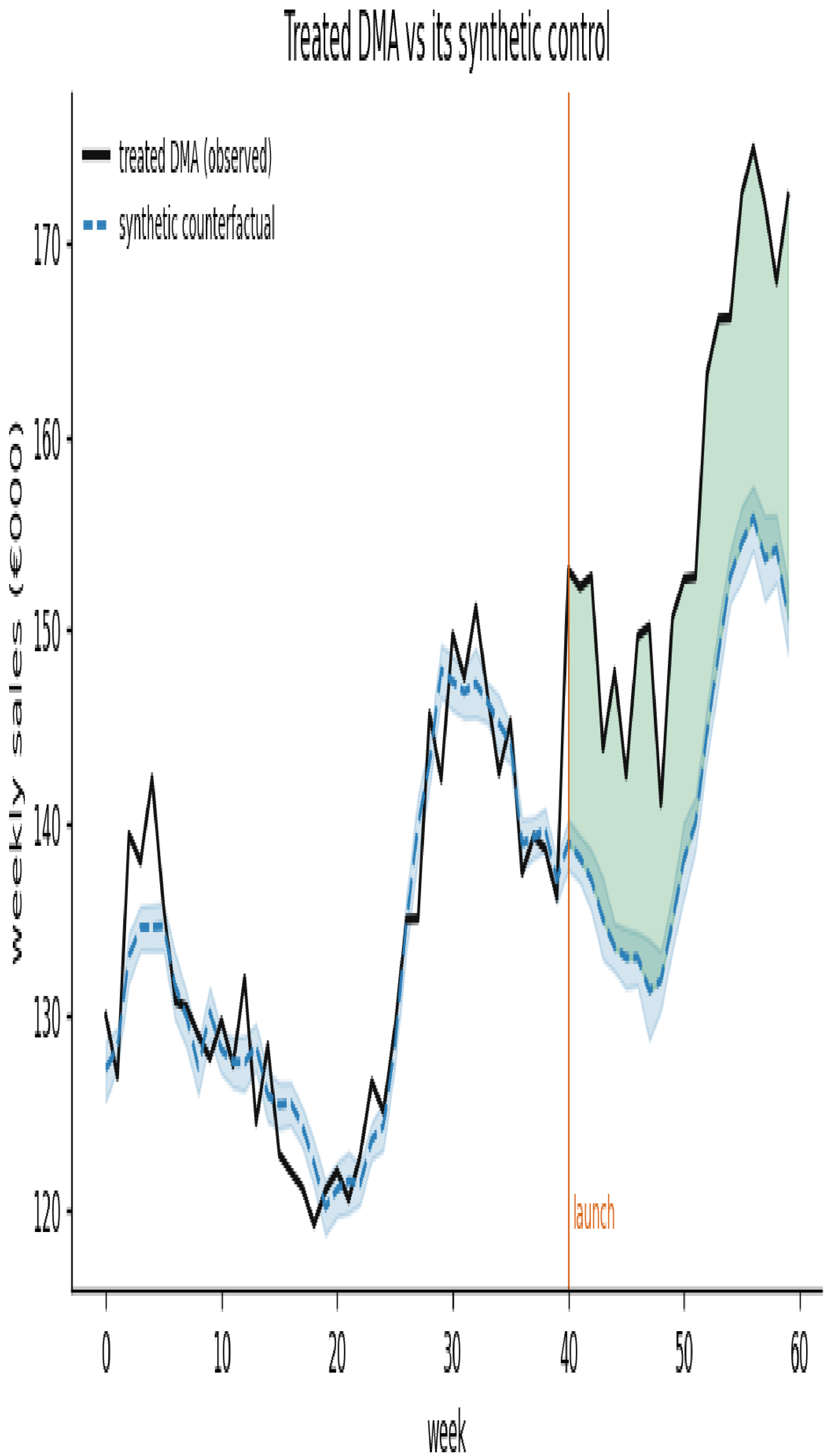

Figure 1: The treated market against its synthetic control. The blend of untreated markets tracks the treated one closely *before* launch (the identifying evidence), then the two diverge after — the green wedge is the estimated lift. Shading is the 90% credible band on the counterfactual.

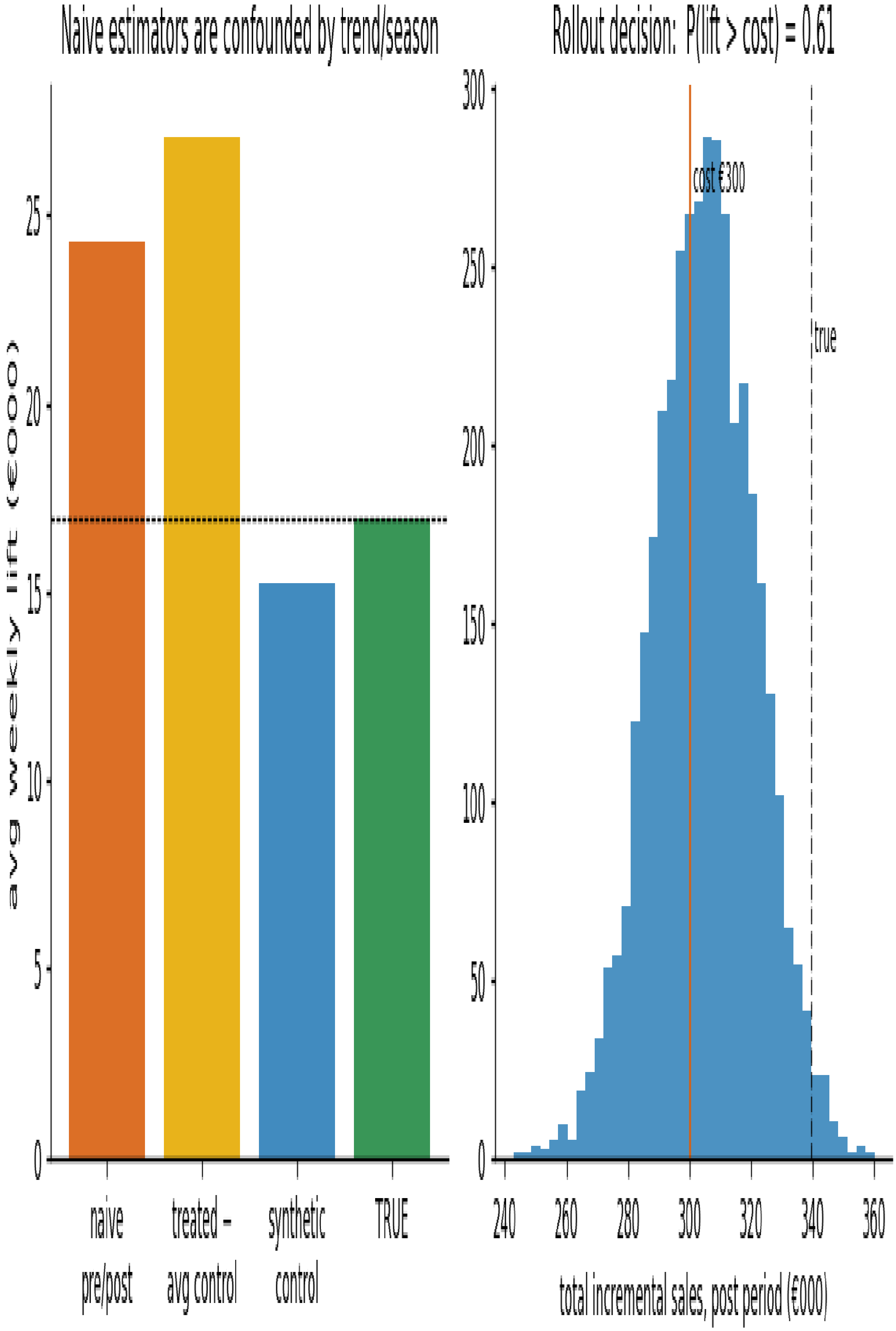

Figure 2: Left: average weekly lift, four ways. The two naive estimators (orange/yellow) overstate by ~50% by absorbing trend and seasonality; synthetic control (blue) lands near the truth (green). Right: the euro decision — see Depth 3.

Depth 2 — Identification rigor: is the effect real, or could noise fake it?

With one treated unit there is no standard error in the usual sense; inference comes from **placebo (permutation) tests** — the Abadie method — and they are more convincing to a business audience than any p-value.

Does the synthetic twin actually fit before launch? The pre-launch gap between treated and synthetic has RMSE **€2.9k** — small relative to the €15–17k effect. A good pre-fit is the licence to trust the post-fit; a synthetic control that can't reproduce the pre-period is extrapolating and its "effect" is meaningless.

Placebo-in-space: would a lift this big appear by chance? We re-run the entire procedure pretending *each untreated market* was the treated one, building its synthetic control from the others. This gives a distribution of "effects" that are known to be null — pure noise and model error. If the real treated market's gap is unremarkable against that distribution, we have nothing; if it stands out, the effect is real.

The treated market's post-launch gap (**€15.9k**) is the **most extreme of 19 valid placebo markets** — i.e., the strongest result the permutation can give, a permutation **p = 0.05**. Read plainly: *when we fake-treated markets that got no campaign, none produced a divergence this large*, so the treated one is unlikely to be noise. (We discard placebos whose pre-fit is much worse than the treated one's, exactly as Abadie prescribes — a badly-fit placebo's "effect" is meaningless.)

Placebo-in-time: is the method inventing effects? We move the launch date *earlier*, to a week before the real campaign, and re-estimate over the still-untreated interval. It finds **€0.9k** — essentially zero. Had it reported a large "effect" before anything happened, the whole design would be suspect. It doesn't.

The residual assumptions worth stating aloud to stakeholders — untestable, like all identification — are **no anticipation** (customers didn't react before launch), **no spillover** (the campaign didn't leak into the control markets), and **overlap** (the treated market's pre-period lies within the range the donors can reconstruct, which the RMSE confirms).

Depth 3 — The decision, in euros

Cumulating the weekly effect over the 20-week post period gives the total incremental sales, *with a posterior distribution* — which is what turns "did it work?" into an actual decision.

- **Total incremental sales: €305k** (90% credible interval **€277k–€332k**). The true value was €339k, sitting just above the interval — the synthetic control is **mildly conservative** here (imperfect weights, a finite pre-period), a known tendency worth flagging rather than hiding.
- **Campaign cost: €300k.** So the point estimate barely clears cost.
- **P(campaign was profitable) = 0.61.** Expected ROI ≈ **+€5k** — essentially break-even.

This is the honest, decision-shaped answer, and it is *not* "yes." The campaign probably paid for itself, but there is a **~40% chance it did not** — so rolling out nationally on this evidence is premature. The rational next steps fall straight out of the numbers: **extend the test window** (more post-weeks tighten the interval), **renegotiate the campaign cost** (the break-even is right at €300k), or **run the burst in a few more markets** to sharpen the estimate before committing the national budget.

Contrast the naive path: the pre/post estimate of €24.3k/wk implies roughly **€486k** of total lift — a confident **green light** to spend nationally. The confounding doesn't just add error; it **flips the decision** from "roughly break-even, wait for more evidence" to "obvious yes." That reversal, and the €300k+ of national budget riding on it, is the entire reason to do this properly.

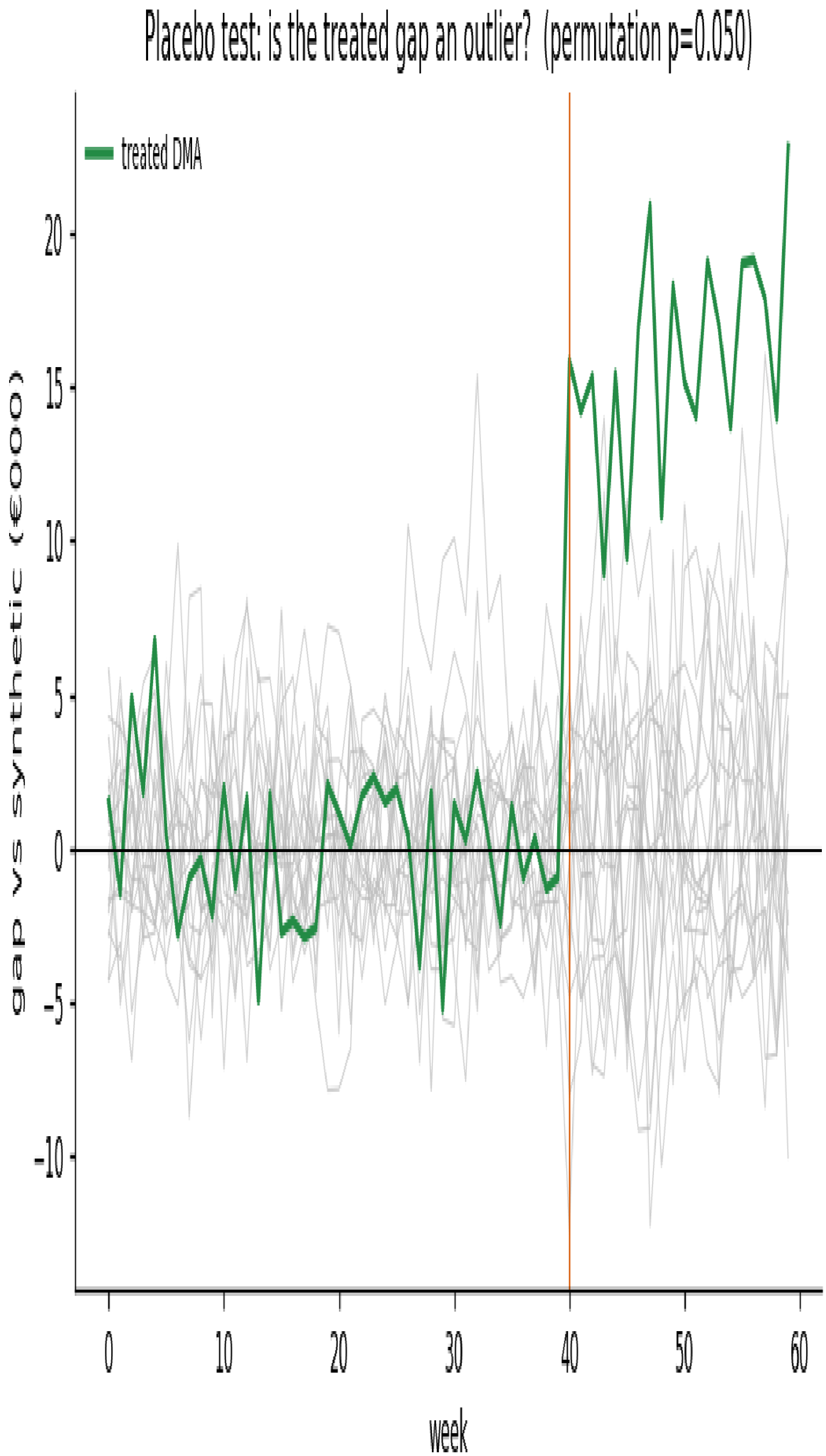

Figure 3: Every grey line is a placebo market’s gap (fake treatment, true effect = 0); the green line is the real treated market. Its post-launch divergence is the single largest of all — the smallest achievable permutation p with this many placebos.

What transfers to the lecture

- **In marketing, the quasi-experiment is the default**, not the exception — most "did X work?" questions arrive with no control group, and before/after is confounded by everything that trends.
- **Synthetic control manufactures the missing control group** from the units you didn't touch, and its credibility rests on a visible, checkable pre-period fit — a picture a CMO can read directly.
- **Placebo tests are the inference**, and they persuade non-technical audiences better than any standard error: *"when we pretended untreated regions got the campaign, none showed a jump this big."*
- **The decision is a probability, not a point** — P(profitable) = 0.61 on a €300k call is a different business conversation than a naive "€486k, ship it," and getting that conversation right is worth far more than the modelling itself.